

DAY 05 – ETHERNET & MAC

Student Name: Rishabh

Course: BCA (Cyber Security)

Day: Day 05

1. What is Ethernet?

Ethernet is a wired networking technology used to connect devices in a Local Area Network (LAN).

2. What is an Ethernet frame?

An Ethernet frame is a unit of data transmitted at the Data Link Layer. It contains source MAC address, destination MAC address, payload and error-checking information.

3. What is a MAC address?

A MAC (Media Access Control) address is a unique 48-bit hardware/network-interface address used for communication on a local network.

4. What does MAC stand for?

MAC stands for Media Access Control.

5. Difference between a MAC address and an IP address?

A MAC address is mainly used for local Layer 2 communication and is associated with a network interface, while an IP address is a logical Layer 3 address used for communication across networks.

6. What is a MAC Address Table?

A MAC Address Table is maintained by a network switch and maps MAC addresses to switch ports.

7. How does a switch learn MAC addresses?

A switch examines the source MAC address of incoming Ethernet frames and records it with the port on which the frame arrived.

8. What is a source MAC address?

The source MAC address identifies the network interface/device that sent the Ethernet frame.

9. What is a destination MAC address?

The destination MAC address identifies the network interface/device to which the Ethernet frame is being delivered on the local network.

10. What is an unknown unicast frame?

An unknown unicast frame is a frame whose destination MAC address is not currently present in the switch's MAC address table.

11. What is flooding?

Flooding is when a switch forwards an unknown-unicast frame out of multiple appropriate ports, except the port on which it was received.

12. What is known unicast forwarding?

Known unicast forwarding occurs when a switch knows the destination MAC address and sends the frame only through the corresponding port.

13. At which OSI layer does Ethernet switching primarily operate?

Ethernet switching primarily operates at Layer 2 – the Data Link Layer.

14. Why does a switch maintain a MAC Address Table?

A switch maintains a MAC address table to efficiently forward frames to the correct port instead of sending every frame to all ports.

15. Why is Ethernet knowledge important in cybersecurity?

It helps cybersecurity professionals understand MAC addresses, switching, ARP behavior and network traffic, and assists in detecting suspicious devices and attacks such as ARP spoofing.

PRACTICAL IDENTIFICATION

Network Adapter Name:- Wi-Fi

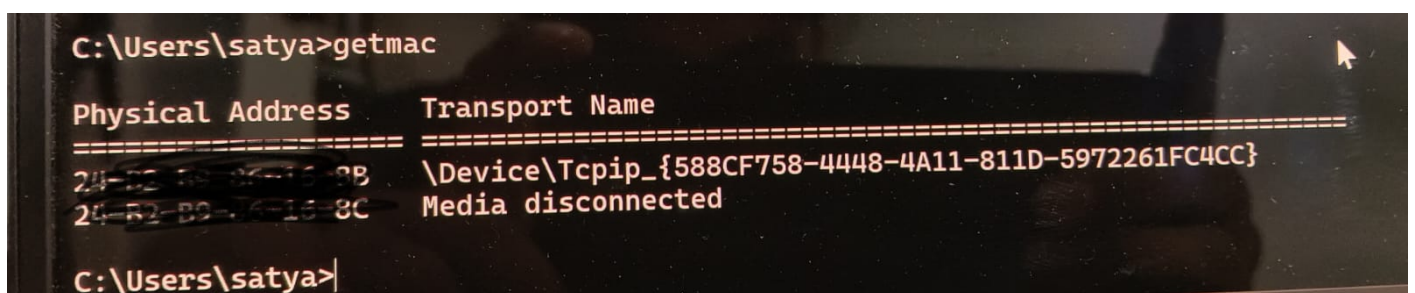
MAC address:- 24-B2-B9-86-16-8B

IPv4:- 10.83.207.207

Default Gateway:- 10.83.207.244

SCREENSHOTS:-

Getmac



```
C:\Users\satya>getmac

Physical Address      Transport Name
=====
24-B2-B9-86-16-8B    \Device\NPF{588CF758-4448-4A11-811D-5972261FC4CC}
24-B2-B9-86-16-8C    Media disconnected

C:\Users\satya>
```

Ipconfig /all

```
C:\Users\satya> ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : Satyam
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Wireless LAN adapter Wi-Fi 3:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8922AE WiFi 7 PCI-E NIC #3
Physical Address. . . . . : 26-B2-B9-86-16-8B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8922AE WiFi 7 PCI-E NIC
Physical Address. . . . . : 24-B2-B9-86-16-8B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2409:40d4:117:eddb:9cbb:94ba:914:8e87(Preferred)
IPv6 Address. . . . . : 2409:40d4:3119:5ca7:c300:564a:2ac4:8da7(Preferred)
Temporary IPv6 Address. . . . . : 2409:40d4:117:eddb:a4ca:4f9c:627e:522e(Preferred)
Temporary IPv6 Address. . . . . : 2409:40d4:3119:5ca7:a4ca:4f9c:627e:522e(Preferred)
Link-Local IPv6 Address . . . . . : fe80::7919:e1e4:a5d3:a82d%8(Preferred)
IPv4 Address. . . . . : 10.83.207.182(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 10 September 2026 12:50:39
Lease Expires . . . . . : 10 September 2026 13:50:37
Default Gateway . . . . . : fe80::415:5dff:fe49:3f44%8
                             10.83.207.244
DHCP Server . . . . . : 10.83.207.244
DHCPv6 IAID . . . . . : 136622777
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-1C-E9-81-D4-AE-52-5B-90-3A
DNS Servers . . . . . : 10.83.207.244
                             2409:40d4:3119:5ca7::94
```

Arp -a

```
C:\Users\satya>arp -a
```

```
Interface: 10.83.207.182 --- 0x8
```

Internet Address	Physical Address	Type
10.83.207.244	06-15-5d-49-3f-44	dynamic
10.83.207.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

